A look into a grocery store's data

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What does the data consist of?

- **Scale:** ~3.4M orders, ~32M prior order-product rows, ~50k products
- Distributed across 21 departments, and 134 aisles

Case Overview

Given the order data (~3.4M orders, ~50k products), I wanted to understand the shape of customer behavior: what gets bought, when, and what makes people come back for the same items.

Questions I started with:

- When do different categories get ordered?
- How loyal are customers to specific products/categories?
- Does cart position say anything about an item?
- Does how often someone shops affect how much they buy per trip?

Cached dataset found, using that

	order_id	product_id	add_to_cart_order	reordered	user_id	order_nu
0	2	33120	1	1	202279	
1	2	28985	2	1	202279	
2	2	9327	3	0	202279	
3	2	45918	4	1	202279	
4	2	30035	5	0	202279	
...	...	...	...	...	...	
33819101	3421063	14233	3	1	169679	
33819102	3421063	35548	4	1	169679	
33819103	3421070	35951	1	1	139822	
33819104	3421070	16953	2	1	139822	

	order_id	product_id	add_to_cart_order	reordered	user_id	order_nu
33819105	3421070	4724	3	1	139822	

33819106 rows × 14 columns

Know your customers

Before anything else, who are we even looking at?

How Many Orders Does a Typical Customer Place?

	order_id
count	206209.000000
mean	16.226658
std	16.662238
min	3.000000
25%	6.000000
50%	10.000000
75%	20.000000
max	100.000000

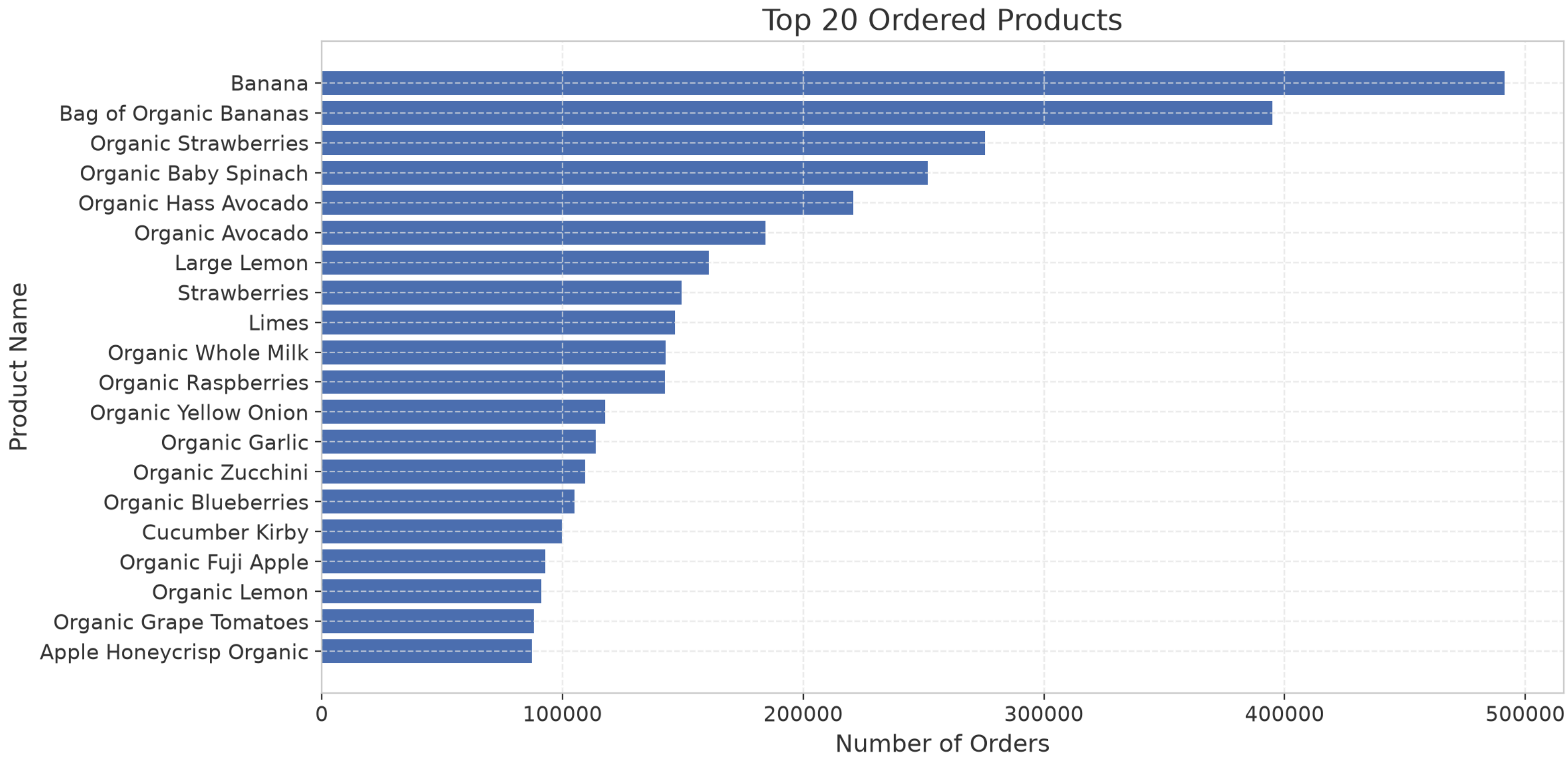
We now know that:

- We have a total of 206,209 customers
- The minimum number of orders for a customer was 3
- The maximum number of orders was 100
- The average customer places 16 orders

What’s In The Cart

Before timing or loyalty, what does this store actually sell, and what do people actually buy?

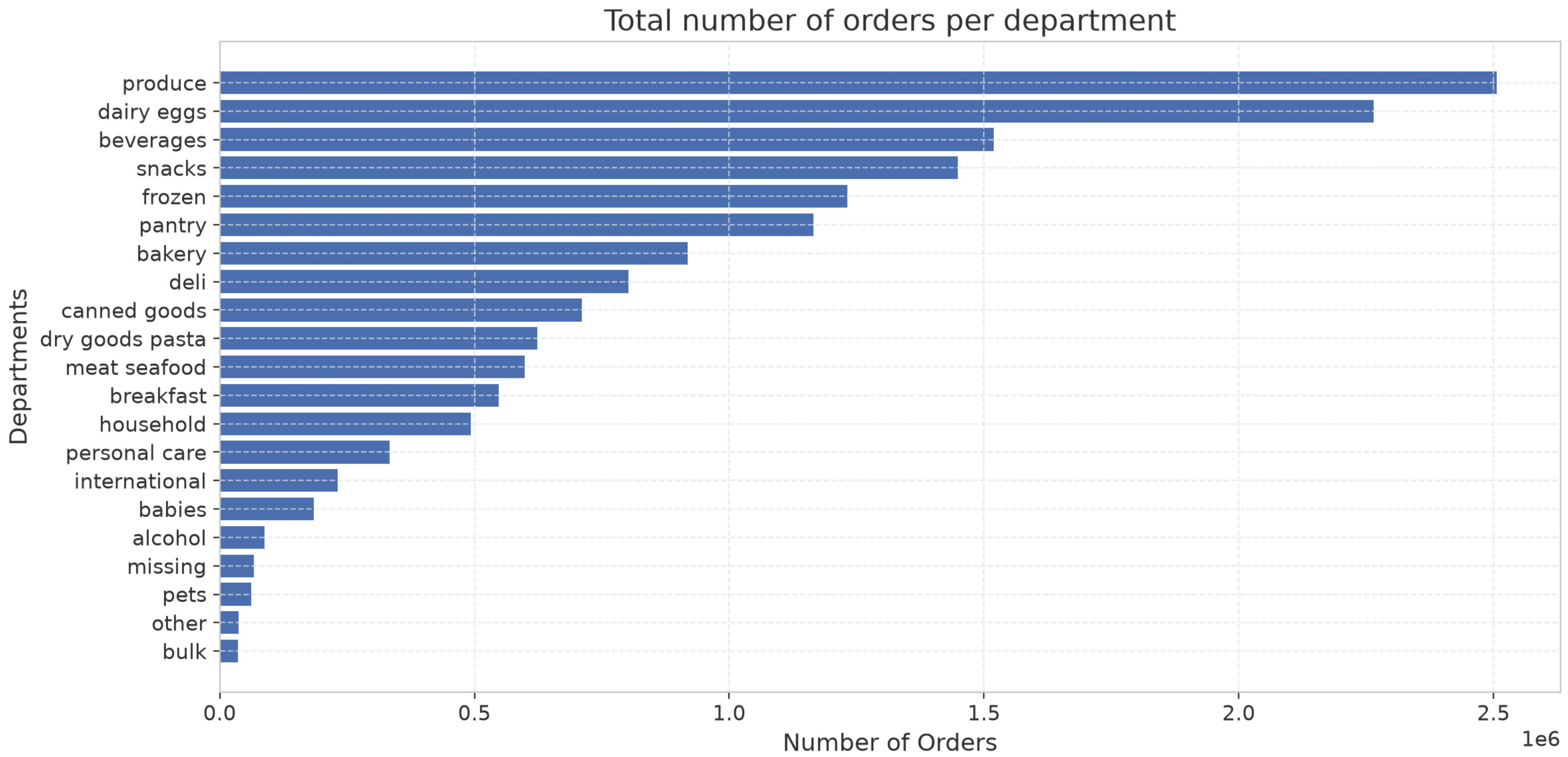
Most Popular Products


Top 20 Ordered Products
Product Name
Banana
Bag of Organic Bananas
Organic Strawberries
Organic Baby Spinach
Organic Hass Avocado
Organic Avocado
Large Lemon
Strawberries
Limes
Organic Whole Milk
Organic Raspberries
Organic Yellow Onion
Organic Garlic
Organic Zucchini
Organic Blueberries
Cucumber Kirby
Organic Fuji Apple
Organic Lemon
Organic Grape Tomatoes
Apple Honeycrisp Organic
0
100000
200000
300000
400000
500000
Number of Orders

We now know that:

- Vegetables and fruits dominate the top order.
- Bananas are the most ordred items on the list. With ~850 thousand orders.
- Spinach is the most ordered vegetable
- Whole milk is the most ordered dairy product

Orders by Department

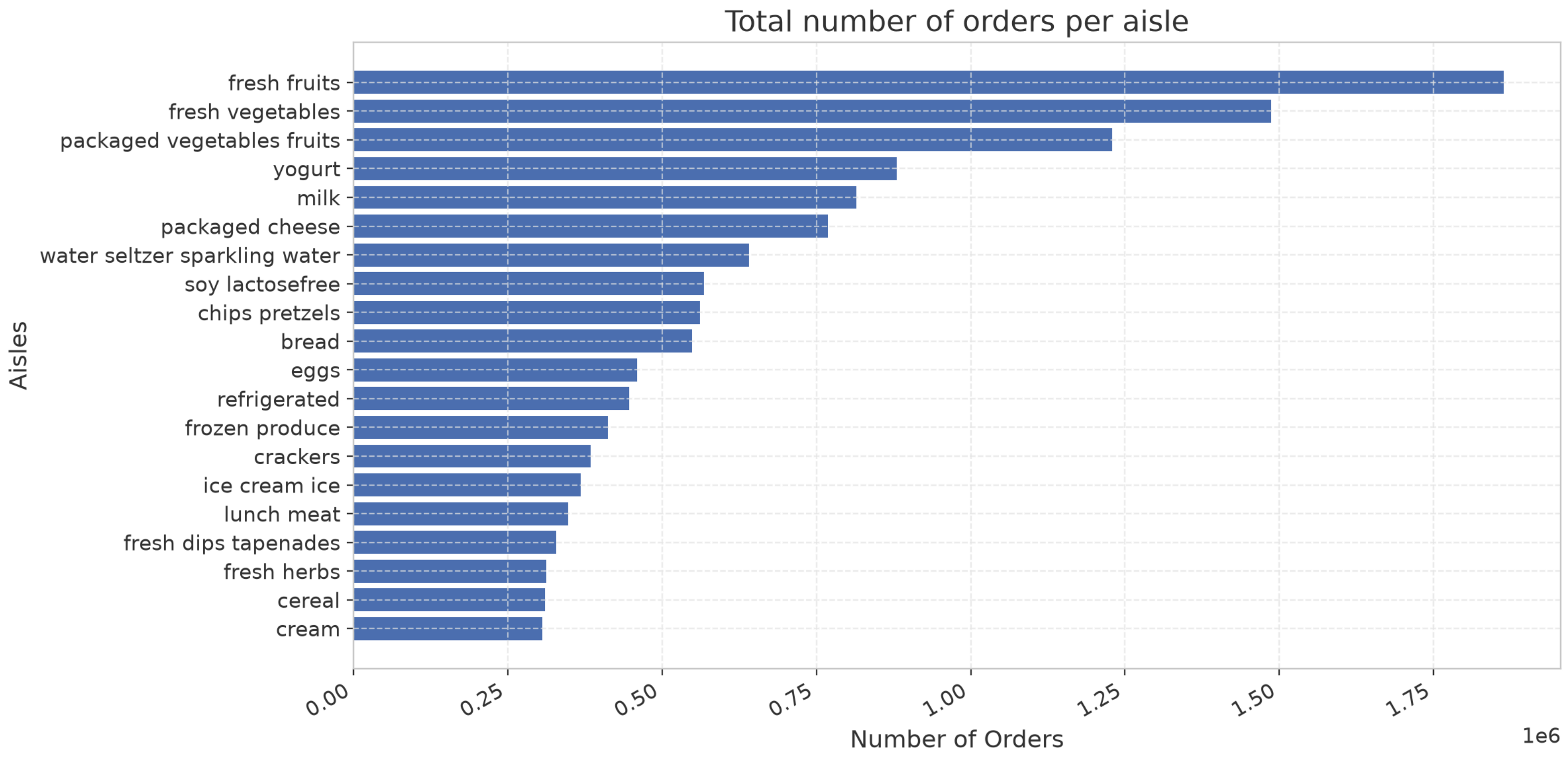


Total number of orders per department
produce
dairy eggs
beverages
snacks
frozen
pantry
bakery
deli
canned goods
dry goods pasta
meat seafood
breakfast
household
personal care
international
babies
alcohol
missing
pets
other
bulk
Departments
0.0
0.5
1.0
1.5
2.0
2.5
1e6
Number of Orders

We now know that:

- The diary & Eggs and the produce departments are the top 2 departments. Which aligns with our findings in the top products section.
- A sudden drop happens at the number of orders in the rest of the departments
- The worst performing departments are: bulk, pets, and other

Orders by Aisle (Top 20)

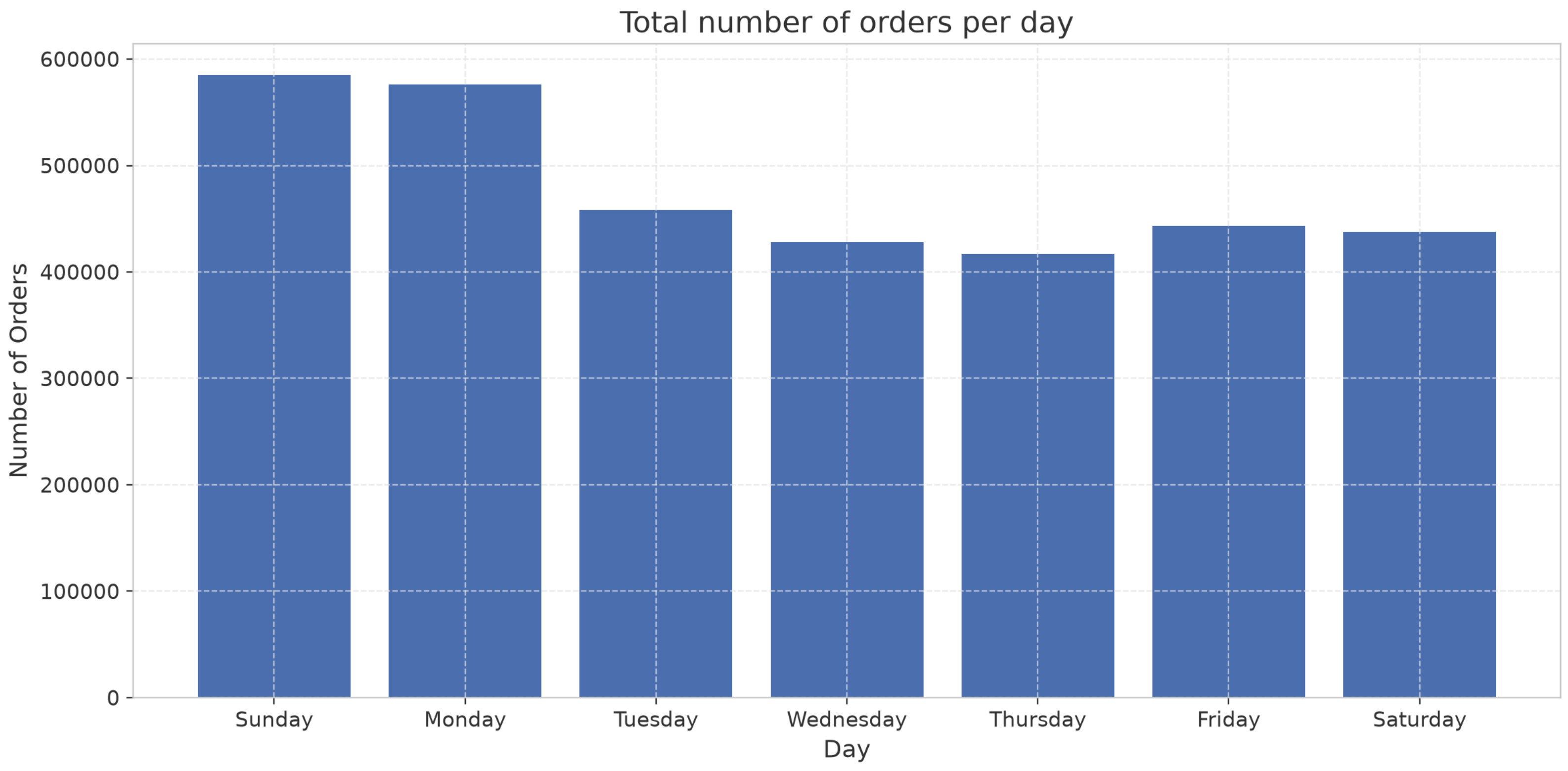

Total number of orders per aisle
fresh fruits
fresh vegetables
packaged vegetables fruits
yogurt
milk
packaged cheese
water seltzer sparkling water
soy lactosefree
chips pretzels
bread
eggs
refrigerated
frozen produce
crackers
ice cream ice
lunch meat
fresh dips tapenades
fresh herbs
cereal
cream
Aisles
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
1e6
Number of Orders

When People Shop

Now that we know what's popular. when does it actually get bought?

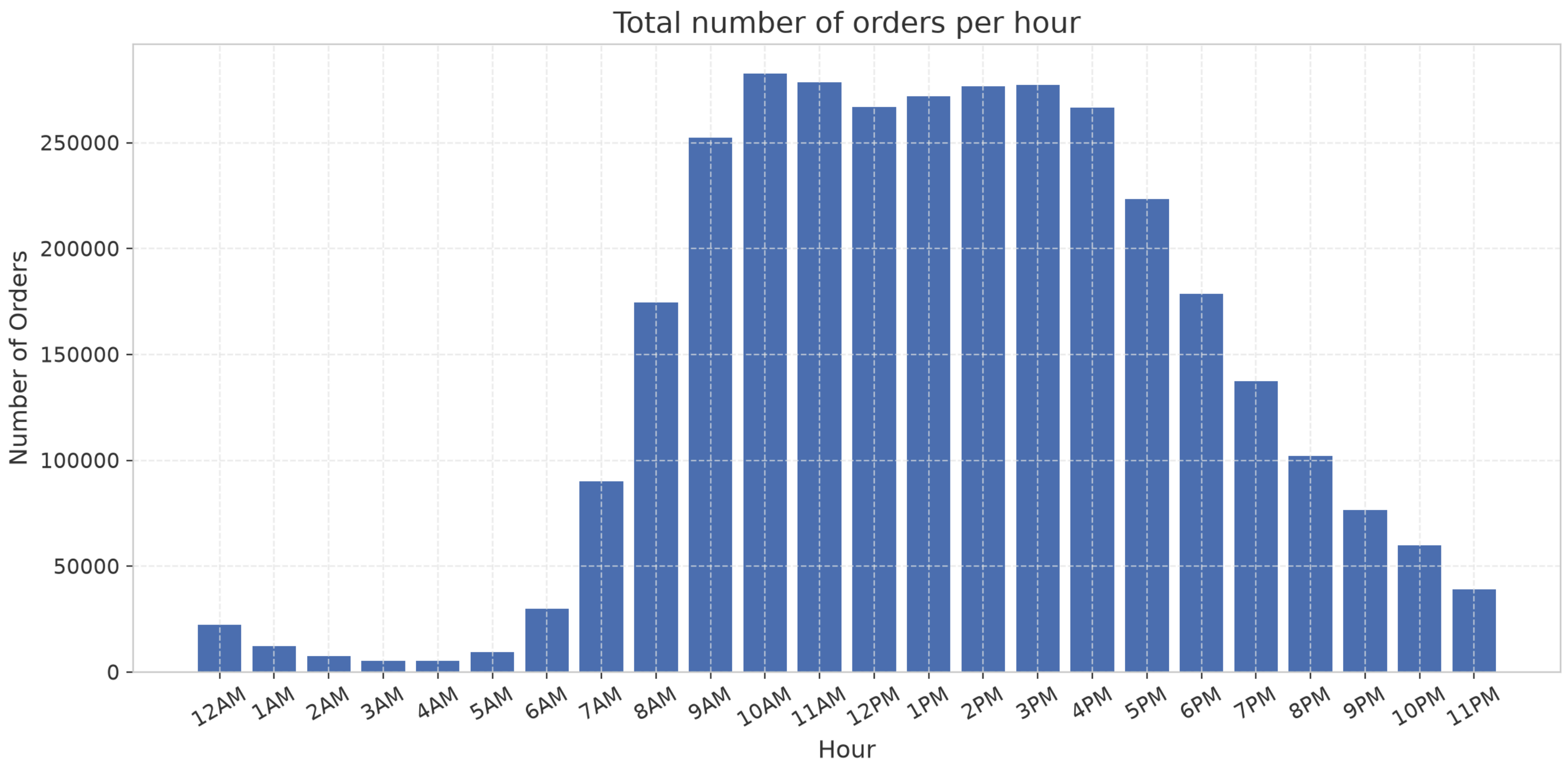
Orders by Day of the Week

	0
Sunday	585237
Monday	576377
Tuesday	458074
Wednesday	428087
Thursday	417171
Friday	443388
Saturday	437749

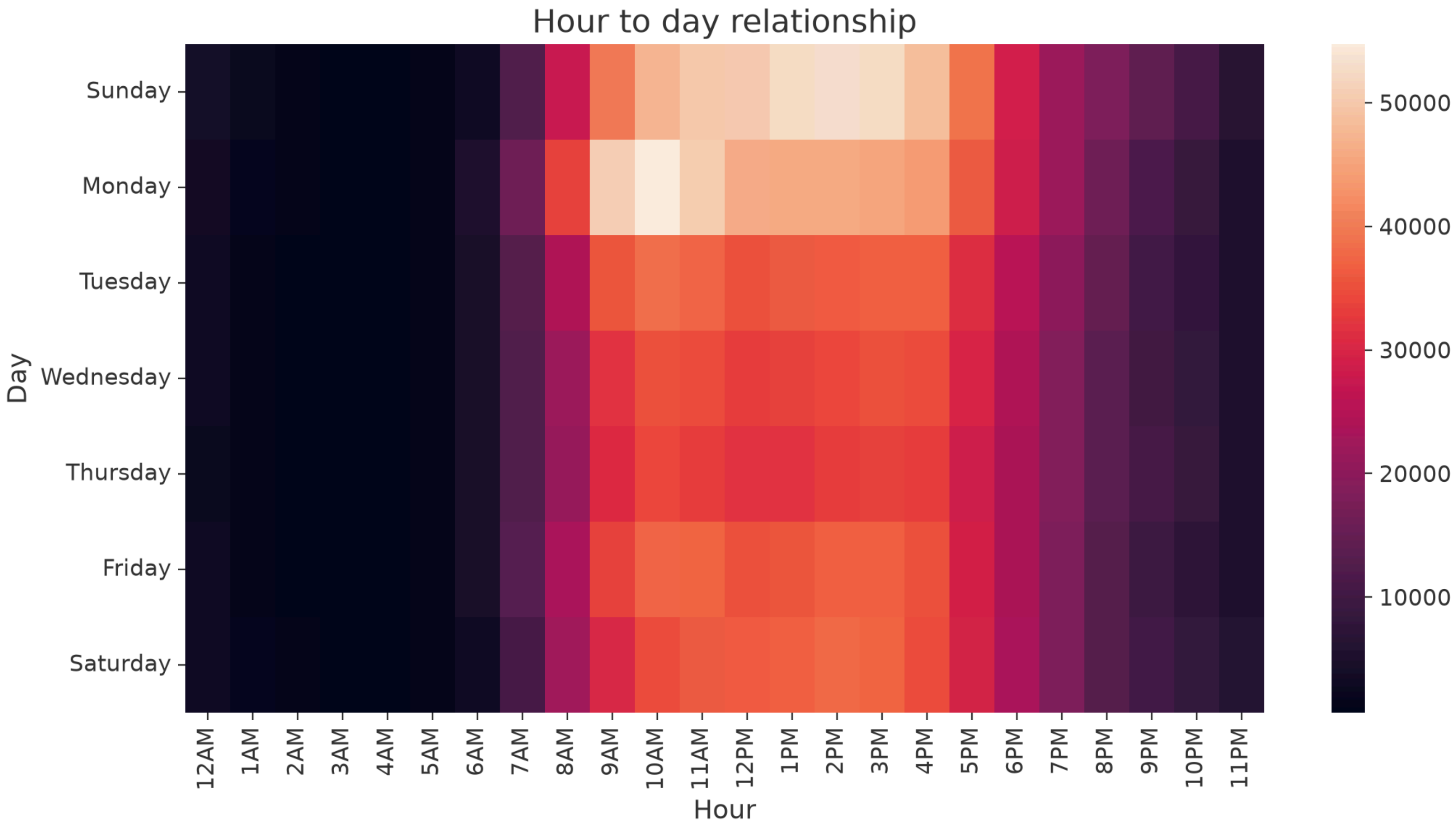


Total number of orders per day
Number of Orders
0
100000
200000
300000
400000
500000
600000
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Day

Orders by Hour of the Day


Total number of orders per hour
Number of Orders
0
50000
100000
150000
200000
250000
12AM
1AM
2AM
3AM
4AM
5AM
6AM
7AM
8AM
9AM
10AM
11AM
12PM
1PM
2PM
3PM
4PM
5PM
6PM
7PM
8PM
9PM
10PM
11PM
Hour

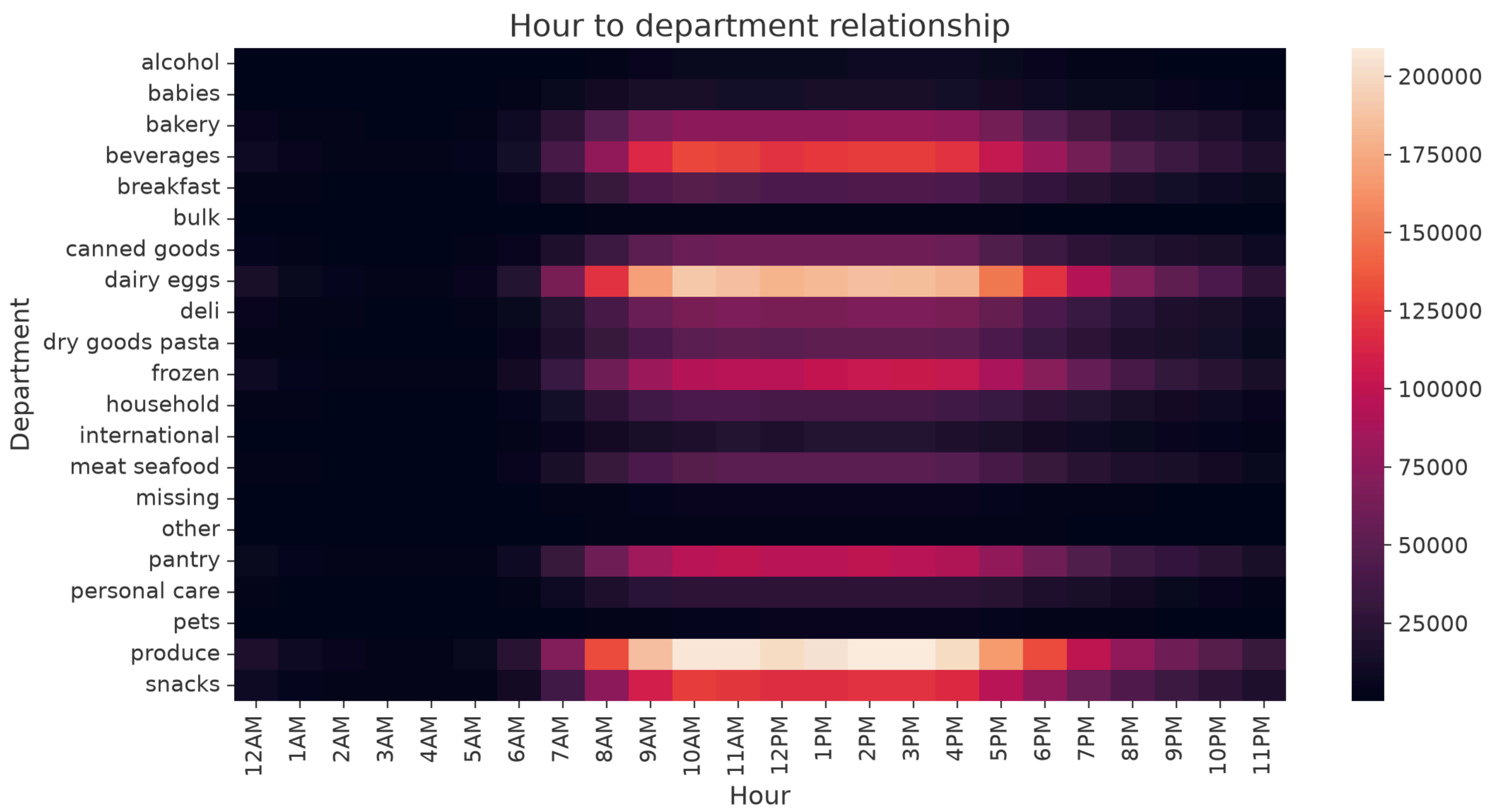
Order Volume: Day vs. Hour


Hour to day relationship
Day
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
12AM
1AM
2AM
3AM
4AM
5AM
6AM
7AM
8AM
9AM
10AM
11AM
12PM
1PM
2PM
3PM
4PM
5PM
6PM
7PM
8PM
9PM
10PM
11PM
Hour
50000
40000
30000
20000
10000

We now know that:

- We get the most orders on Sundays and Mondays
- We get the least orders on Thursdays
- The least amount of orders happens between midnight and 6 AM
- The busiest part of the day is from 10 AM to 4 PM. With 10 AM being the busiest.
- We get the most amount of orders on Mondays at 10 AM

Order Timing by Department


Hour to department relationship
Department
alcohol
babies
bakery
beverages
breakfast
bulk
canned goods
dairy eggs
deli
dry goods pasta
frozen
household
international
meat seafood
missing
other
pantry
personal care
pets
produce
snacks
12AM
1AM
2AM
3AM
4AM
5AM
6AM
7AM
8AM
9AM
10AM
11AM
12PM
1PM
2PM
3PM
4PM
5PM
6PM
7PM
8PM
9PM
10PM
11PM
Hour
200000
175000
150000
125000
100000
75000
50000
25000

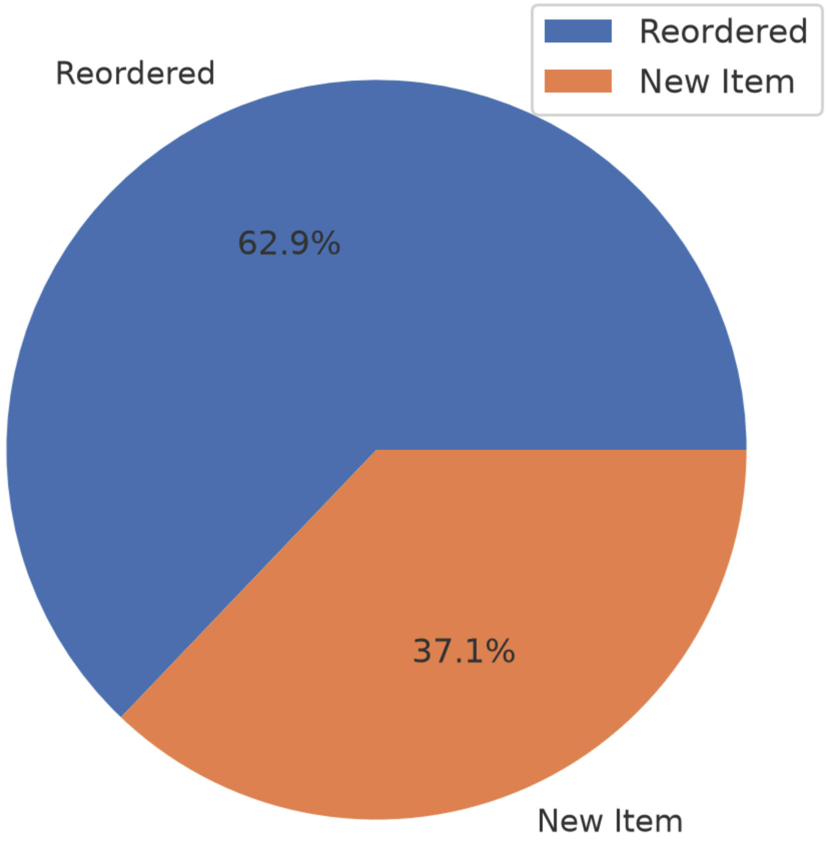
Takeaway: Timing x Department

- The pattern from our previous findings presists. all of the departments get the most orders at day.

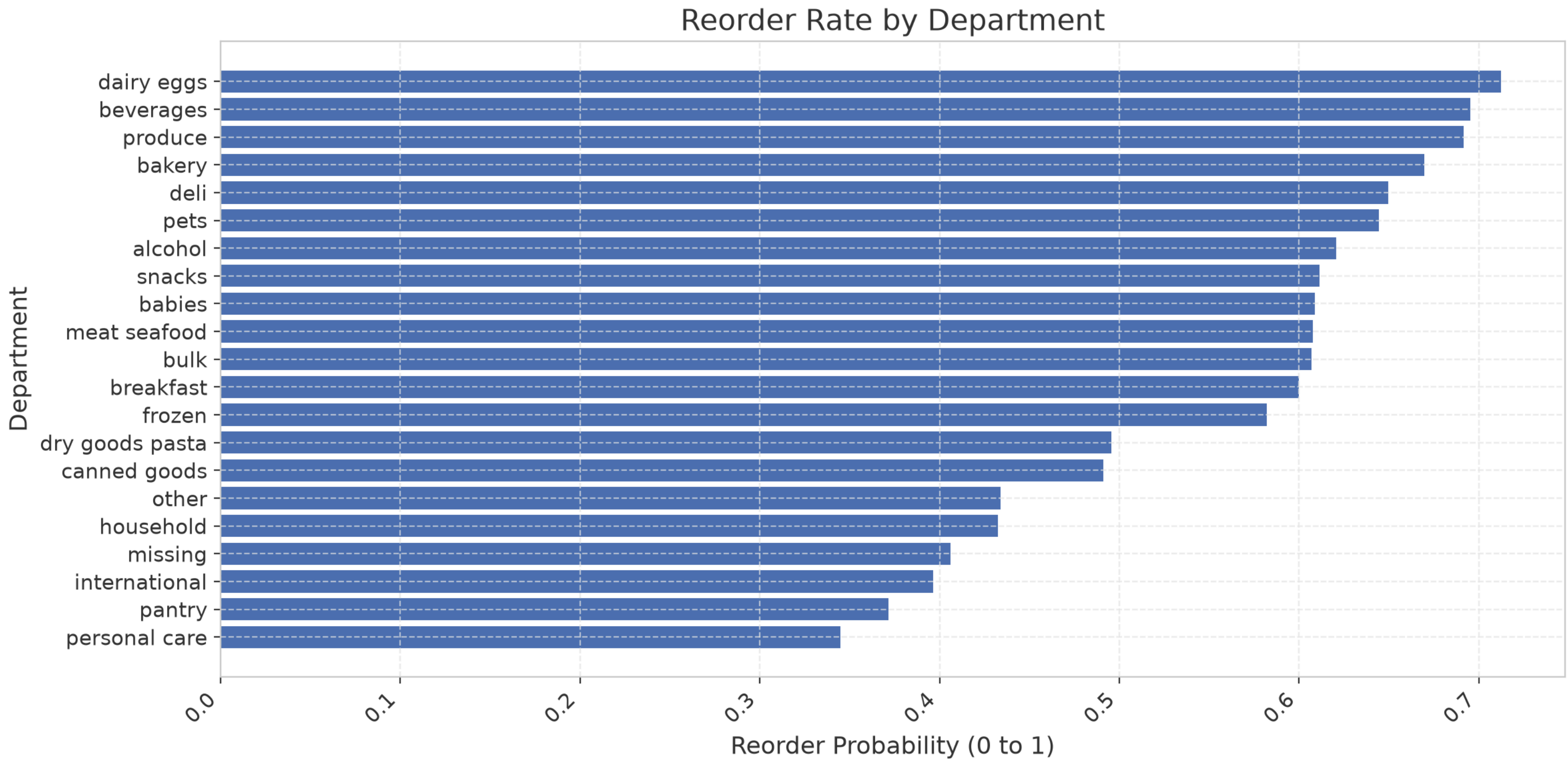
Loyalty & Reorders

Who keeps coming back for the same thing, and what?

Overall Reorder Rate


Reordered
New Item
Reordered
62.9%
37.1%
New Item

Reorder Rate by Department

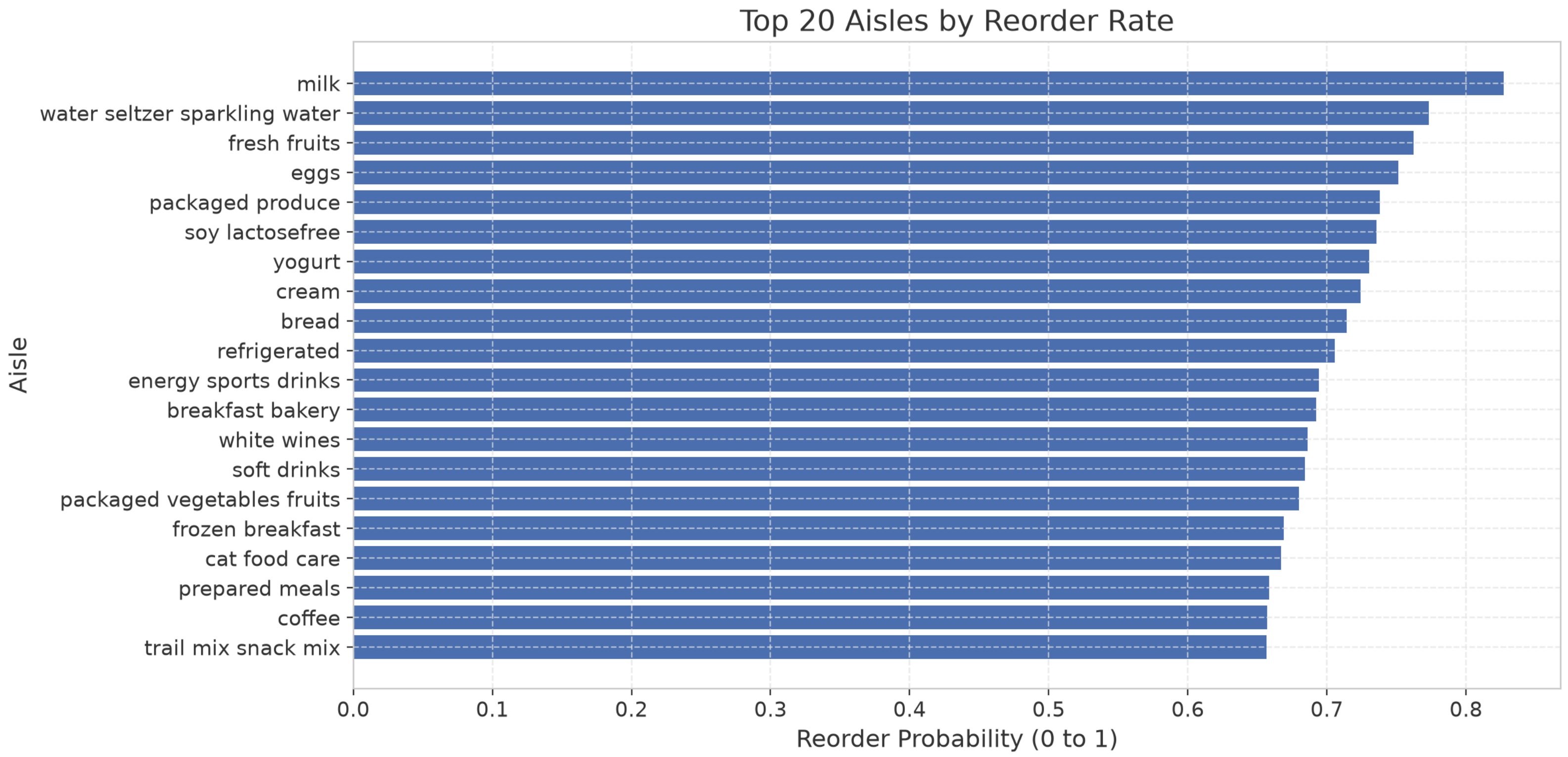


Reorder Rate by Department
dairy eggs
beverages
produce
bakery
deli
pets
alcohol
snacks
babies
meat seafood
bulk
breakfast
frozen
dry goods pasta
canned goods
other
household
missing
international
pantry
personal care
Department
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Reorder Probability (0 to 1)

We now know that:

- The reorder rate is 62.9%
- Diary and eggs have the top reorder rate
- Personal care and pantry have the lowest reorder rate

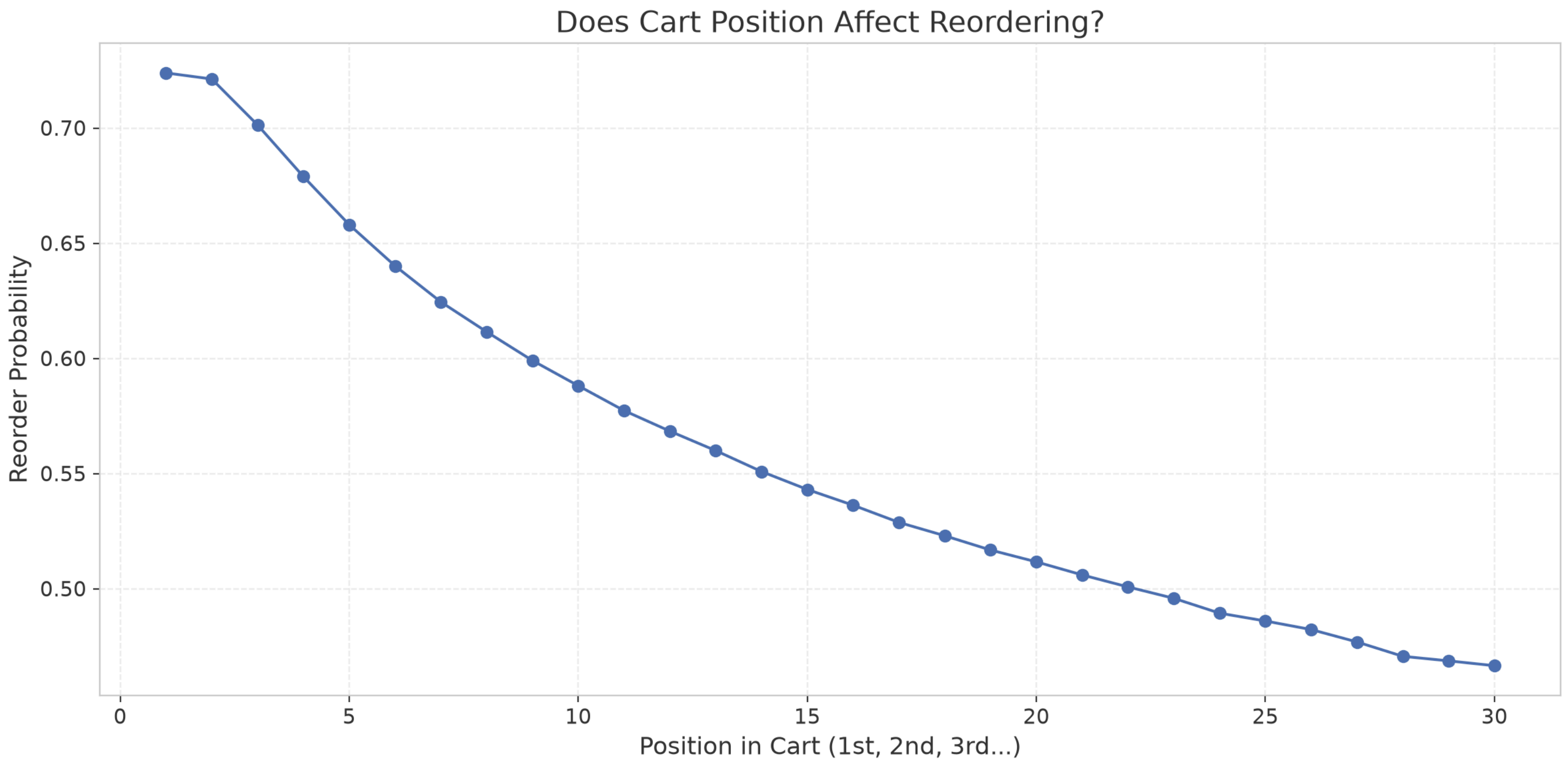
Reorder Rate by Aisle (Top 20)


Top 20 Aisles by Reorder Rate
milk
water seltzer sparkling water
fresh fruits
eggs
packaged produce
soy lactosefree
yogurt
cream
bread
refrigerated
energy sports drinks
breakfast bakery
white wines
soft drinks
packaged vegetables fruits
frozen breakfast
cat food care
prepared meals
coffee
trail mix snack mix
Aisle
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Reorder Probability (0 to 1)

Cart Behavior

Does where an item lands in the cart say anything about it?

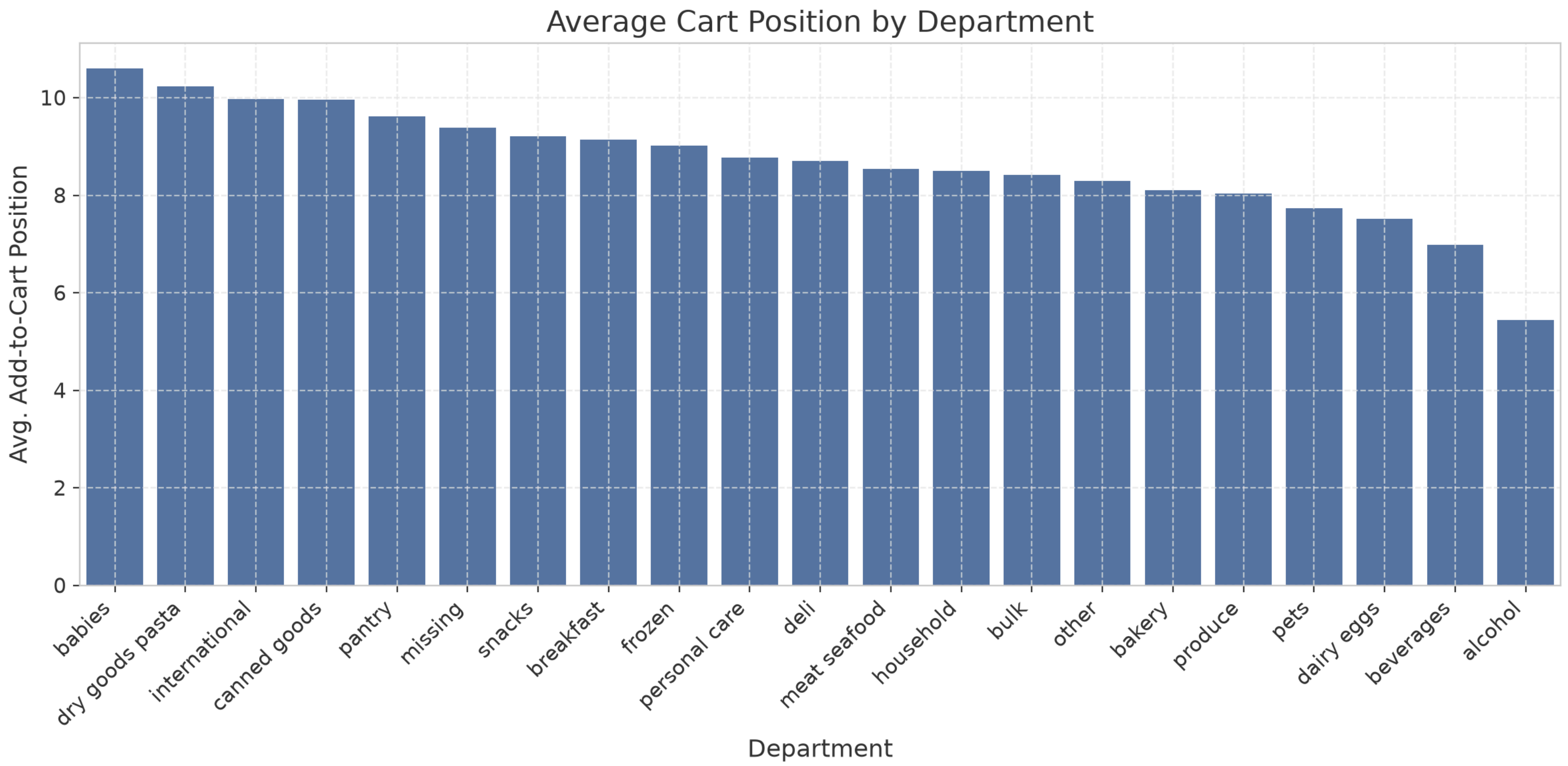
Cart Position vs. Reorder Probability


Does Cart Position Affect Reordering?
Reorder Probability
0.70
0.65
0.60
0.55
0.50
0
5
10
15
20
25
30
Position in Cart (1st, 2nd, 3rd...)

We now know that:

- Reorder rate is directly affected by cart position
- We see a steady decrease in reorder rate as the position in cart increases

Cart Position by Department



Average Cart Position by Department
Avg. Add-to-Cart Position
Department
0
2
4
6
8
10
babies
dry goods pasta
international
canned goods
pantry
missing
snacks
breakfast
frozen
personal care
deli
meat seafood
household
bulk
other
bakery
produce
pets
dairy eggs
beverages
alcohol

We now know that:

- Alcohol and beverages get placed earlier than other products
- Babies department gets bought at last most of the time

Cadence & Basket Size

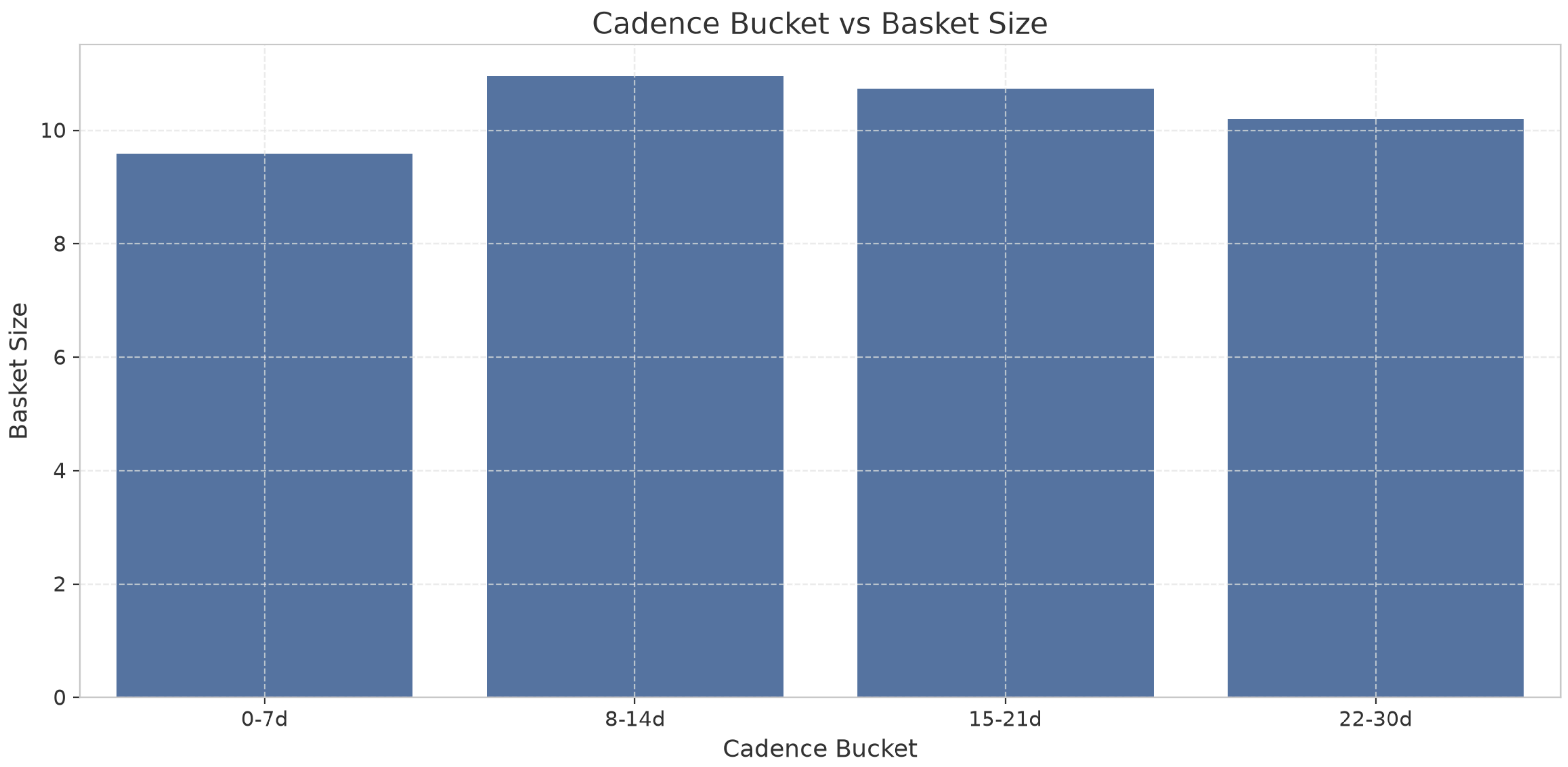
How often people come back, and whether that changes how much they buy each time.

Average Time Between Orders

10.972287422998502

Basket Size by Shopping Frequency

	basket_size
cadence_bucket	
0-7d	9.581107
8-14d	10.959508
15-21d	10.735771
22-30d	10.193979


Cadence Bucket vs Basket Size
Basket Size
0
2
4
6
8
10
0-7d
8-14d
15-21d
22-30d
Cadence Bucket

We now know that:

- infrequent shoppers buy more products compared to other people. Though the difference isn't that big

Summary

- Loyalty is high but concentrated: 62.9% of items are reorders overall, but that loyalty lives almost entirely in dairy/eggs and produce. personal care and pantry on the other hand barely get repeat purchases.
- Shopping timing is consistent: the same daytime pattern (peak 10 AM–4 PM, quiet overnight, Sunday/Monday busiest, Thursday quietest) holds across every department. No category has its own rhythm.

- Cart order reflects the customers' thougts: alcohol and beverages get added early (planned purchases) while baby products get added last (impulse or secondary trip items).
- Infrequent shoppers buy slightly more per trip than frequent ones. the gap is real but small.

Next Steps

- **Promote on Thursdays.** It's the lowest-order day of the week and a targeted promotion there is guaranteed to do better, compared to an already good day like Sunday where it might not change much.
- **Drive repeat purchases in personal care and pantry.** These have the store's lowest reorder rates despite reasonable order volume. We should increase the quality of our products there and put more points towards them in the loyalty system (If we have one)

- **Reposition low-traffic departments (bulk, pets, other) with bundle offers** rather than straight discounts. These departments aren't getting browsed in the first place, so the goal is getting them into the cart at all, not just cheaper.

- **Use cart-position data for placement.** Items added late (baby products) are strong candidates for placement near the cashier, since they're already being treated as an afterthought. Putting them somewhere visible might increase their order rates more than fighting the user habbits

- **Basket affinity analysis** We should check what products get bought together.

Thank You!